

Mini_Project_2_StageIII

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Libraries

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(rvest)
```

Attaching package: 'rvest'

The following object is masked from 'package:readr':

guess_encoding

```
library(polite)
library(tidyverse)
library(stringr)
```

Starting Up and Getting URLs

```
# Step 0: Check if access works
robotstxt::paths_allowed("https://www.blm.gov/visit/search?field_location=100035&field_activi
```

```
www.blm.gov
```

```
[1] TRUE
```

```
# TRUE
```

```
# This scrapes the page for all the links to each BLM location in Colorado
scrape_links <- function(link_pages) { # This takes a search page
  Sys.sleep(5) # This abides by the 5 second delay rule
  session <- bow(link_pages, force = TRUE) # This bows and starts a session
  page <- scrape(session) # This scrapes the session
  BLM_name <- page |> html_elements("a[hreflang='en']") |> html_attr("href") # This extracts
  tibble(link = str_c("https://www.blm.gov", BLM_name))
  # This saves the full link
}
```

```
park_pages <- vector("list", length = 4)
# This says output a list vector with 4 rows
for (i in 0:3) { # This says to run for pages 0-3
  link_pages <- str_c("https://www.blm.gov/visit/search?field_location=100035&field_activiti
# This combines search page with page number
  park_pages[[i + 1]] <- scrape_links(link_pages)
# This crawls each page and applies the scrape links function
}
```

```
df_pages <- bind_rows(park_pages) # Thus binds every vector from park_pages
head(df_pages)
```

```
# A tibble: 6 x 1
```

```
  link
<chr>
```

```
1 https://www.blm.gov/visit/18-road-north-fruita-desert
2 https://www.blm.gov/visit/alpine-loop
3 https://www.blm.gov/visit/animas-forks
```

```
4 https://www.blm.gov/visit/animasgrandviewskyline-rmz
5 https://www.blm.gov/visit/ant-hills-wsa
6 https://www.blm.gov/visit/bangs-backcountry
```

Text Scraping for the Dataset

```
## From 10_web_scraping

# Helper function to reduce html_nodes() |> html_text() code duplication
get_text_from_page <- function(page, css_selector) {
# This looks at each page and the given selector and extracts text
  page |>
    html_nodes(css_selector) |>
    html_text()
}

add_na <- function(x) {
# This takes a string and if it is length = 0, it outputs an NA, or returns the string itself
  if (length(x) == 0) NA else x
}

# Main function to scrape and tidy desired attributes
scrape_page <- function(url) {
  Sys.sleep(5) # reduces lag on website
  session <- bow(url, force = TRUE) # checks in to the website
  page <- scrape(session) # scrape

  title <- get_text_from_page(page, ".-title")
  address_1 <- get_text_from_page(page, ".-address-1")
  address_2 <- get_text_from_page(page, ".-address-2")
  city <- get_text_from_page(page, ".-city")
  state <- get_text_from_page(page, ".-state")
  zip <- get_text_from_page(page, ".-zip-code")
  lat <- get_text_from_page(page, ".views-field-field-latitude")
  long <- get_text_from_page(page, ".views-field-field-longitude")
  phone <- get_text_from_page(page, ".views-field-field-phone")
  description <- get_text_from_page(page, ".layout__region--first > div:nth-child(3) > div:nth-child(1)")
  directions <- get_text_from_page(page, ".-directions")
  activities <- get_text_from_page(page, ".-paragraph-activities")

  tibble(BLM_Name=add_na(title),
```

```

    Address_1= add_na(address_1),
    Address_2= add_na(address_2),
    City= add_na(city),
    State= add_na(state),
    Zip_Code= add_na(zip),
    Latitude = add_na(lat),
    Longitude = add_na(long),
    Phone_Number = add_na(phone),
    Description = add_na(description),
    Directions = add_na(directions),
    Activities = add_na(activities))|>
mutate(BLM_Name = word(BLM_Name, 1, sep = fixed('|')),
# Extracts the word(s) before the | separator
      Activities = str_squish(Activities))
#Removes the \n between activities.
}

# Useful testing frame for the above function
test_page<- scrape_page("https://www.blm.gov/visit/vale-tears-wsa")

# Collects the scraped info for every page/location
park_addresses <- vector("list", length = 151) #outputs a vector
for (i in 1:151) {
  park_addresses[[i]] <- scrape_page(df_pages$link[i])
}

# This makes our dataframe of park information
df_addresses <- bind_rows(park_addresses)

# This removes duplicated rows. Only three locations were effected
df_addresses <- df_addresses|>
  distinct(BLM_Name, .keep_all = TRUE)

head(df_addresses)

# A tibble: 6 x 12
  BLM_Name Address_1 Address_2 City State Zip_Code Latitude Longitude
  <chr>      <chr>      <chr> <chr> <chr> <chr>      <chr>      <chr>
1 "18 Road - North ~ <NA>      <NA>      <NA> <NA> <NA>      39.334      -
108.704
2 "Alpine Loop "    "Gunniso~ "2500 E.~ "Gun~ Colo~ 81230      38.0319      -
107.3119

```

```

3 "Animas Forks " "Gunniso~ "210 Wes~ "Gun~ Colo~ 81230 37.93123 -
107.570~
4 "Animas/Grandview~ "Tres Ri~ "29211 H~ "Dol~ Colo~ 80323 37.22384 -
107.856~
5 "Ant Hills WSA " "455 Eme~ <NA> "Cra~ Colo~ 81625 0 0
6 "Bangs Backcountr~ <NA> <NA> "Gra~ Colo~ <NA> 38.91 -
108.566
# i 4 more variables: Phone_Number <chr>, Description <chr>, Directions <chr>,
# Activities <chr>

```

Making a CSV

```
write.csv(df_addresses, "blm_colorado.csv")
```